

Instructor and Course Information

Course: ASTRO 237W: Astronomy Communication

Instructors:

Name	Email	Office Hours	Office
Cayla Dedrick (they/she)	ckd5370@psu.edu	TBD	Davey 401
Chloe Klare (she/her)	chloe.klare@psu.edu	TBD	Davey 445a

Class location: Patterson Building, Room 304

Meeting times: Tuesday & Thursday, 3:05-4:20pm

Office hours and location: TBD, Additional times are available by appointment (email or talk to us after class to schedule)

Course Description

ASTRO 237W: Astronomy Communication is a course designed to teach students the skills necessary to become successful member of the astronomical community. In this class, you will learn about and practice modes of communication that are used every day in academia: technical writing and reading, presenting to different audiences, designing posters and slides, and interfacing with other scientists and the public. There will be a strong focus on improving scientific writing skills, practiced by submitting small-but-regular assignments inside and outside of class. By the end of this course, students will have many products that can be applied to other areas of their careers: a moderate length technical essay on a topic of their choice, a presentation and a poster on the same topic, and a suite of professional development products.

Course Objectives

After this class, students will be able to...

- Create and format documents using LaTeX in Overleaf.
- Identify the format and purpose of common sections found in journal articles in the physical sciences.
- Critically analyze and improve their own scientific writing with attention to tone, grammar, and format.
- Evaluate, understand, find, and cite academic journal articles.

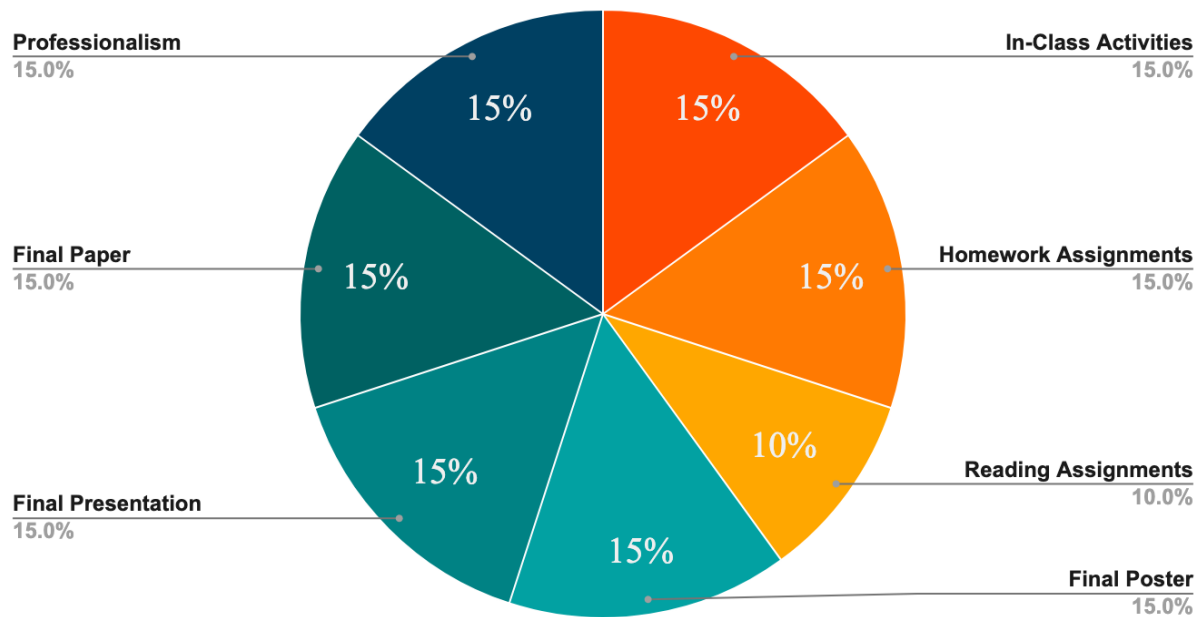
- Understand the collaborative part of the scientific process.
- Present their findings to different audiences in a clear, effective, and engaging manner.
- Apply these skills to REUs, fellowships, graduate school and job applications, and other “asks”
- Communicate effectively with collaborators in order to maintain good working relationships

Class Format

This course will be taught via lecture, collaborative in-class discussion activities, and reading. This will be strengthened by assignments including short presentations of different formats and short written assignments with a focus on peer review and revision.

Grading Policy

In-Class Activities	15%
Homework Assignments	15%
Reading Assignments	10%
Final Poster	15%
Final Presentation	15%
Final Paper	15%
Professionalism	15%



Although grades will not be curved, there will be opportunities for extra credit.

In-class activities:

Many of the class periods will contain in-class activities and discussions which will typically be due by the end of class day (11:59 pm). These assignments will be graded on a three-point scale of insufficient, sort of sufficient, and sufficient and will mainly be used for feedback between students and instructors.

The in-class activities will often include sharing and discussion, either in small groups or to the entire class. Reasonable accommodations or alternative instruction will be provided for students with a persistent reason to decline discussions (anxiety, hearing impairment, speech impediment, etc.) provided they contact the instructor as early as possible. Please contact the instructor if you are unable to work with a specific classmate (i.e., harassment, stalking, or abuse) to avoid being grouped with them.

Homework Assignments:

Homework will be assigned in class and will usually be given at least one week to complete. They should be submitted on Canvas. Homework assignments will include reading, writing, peer editing, and the creation of materials. Ideally, homework should be turned in by the beginning of the class period. These assignments are considered formal writing and will be graded as such. Rubrics for assignments are on their Canvas pages. Homework will be announced in class at least once, but please pay attention to deadlines.

Reading Assignments:

You will be assigned one to two reading assignments each week that are accompanied by a handful of questions. The answers to these questions can be as informal as you would like, but should be properly formatted and at least two sentences per question. These take 10-40 minutes per assignment, depending on the assignment and the student.

Final Project (poster, presentation, paper):

The final poster, presentation, and paper will all be due near the end of the semester (see class schedule). All three of these will cover the same semester-long project, either a small research project or a literature review. The poster and presentation grades will include a grade for peer reviewing other projects, and in turn, a grade given by peers.

Professionalism:

As you move forward in your life and farther from coursework, deadlines and penalties for not reaching them are few and far between. However, communicating your progress towards deadlines and if extensions are necessary is incredibly important. Additionally, interacting with your peers in ways that would encourage future relationships and collaborations is crucial to any career path. For this course, there will be no formal late penalties or attendance policies. This being said, if you know that you will not be able to attend a class for whatever reason or finish an assignment by the posted deadline, then you are responsible for communicating this with your instructors via email before the class/deadline. Failure to do so will result in late penalties (see Late Policy) a lowering of your professionalism grade. You will also lose points on your professionalism grade if your contributions to the classroom environment are not supportive and collaborative.

Extra Credit:

Reading Assignment Extra Credit: Email all instructors examples of really good/bad STEM communication (talks, posters, blogs, papers) along with an explanation about why it is good or bad. One example will be worth a 10% bump to a reading assignment grade, up to 100%.

Project Extra Credit: There are numerous opportunities for talks / public lectures in the astronomy department and other related fields. For 1% extra credit on any final project component (up to 15% across the three components), attend a talk, fill out a rubric and questionnaire on the communication aspects of talk (not the science), and email it to all instructors.

Class Texts

You are not required to purchase any texts for this class, and there will not be a single resource that we follow. We will, however, post short weekly reading assignments.

Attendance

Attendance is not directly graded, however, graded in-class activities will often be assigned and collected at the end of a class period. We highly recommend that you attend class, as there's no textbook or single resource that has all of the information that we'll be covering in the lectures and in-class activities. However, we know that life happens. If you have to miss a class, please send an email to both instructors at least an hour before class with the reason that you can't attend. **Please just be honest!** Physical health, mental health, family/personal obligations, travel etc. can all be more important than a single lecture, and we understand that. So long as you tell us beforehand, you're honest about the reason for your absence, and we don't see any persistent trends of absences, you will not be penalized for missing a class. You will, however, still be expected to complete any in-class activities to receive full-credit.

Late Policy

There are no penalties associated with turning in assignments late in this course. We're all adults who have a lot to deal with and sometimes things happen. If you are unable to turn in an assignment on time, please email both instructors and give us an approximate date by which you can complete the assignment. As long as we receive an email before the deadline, you will not lose any credit. For each day that the assignment is late and that you do not communicate with us, you will lose 10% credit on the assignment, up to a maximum of 50%.

Academic Integrity

We encourage you to work together with your classmates to complete the homework assignments. However, all work must be your own, and plagiarism is strictly not allowed. If you use a source, cite it properly. Plagiarism is not only copying and pasting without using quotations and citing a source, but includes borrowing a source's *structure* (e.g., keeping phrases but rearranging them, replacing words with synonyms). Any student caught plagiarizing (another student or a source) or assisting another student to plagiarize will receive a zero on the assignment on the first infraction. Please see the [Eberly College of Science Academic Integrity page](#) for full details of the policies to which we will abide.

University Resources

Disability Policy: Penn State welcomes students with disabilities into the University's educational programs. If you have a disability-related need for reasonable academic adjustments in this course, contact the Office of Student Disability Resources (SDR) by phone at 814-863-1807 (V/TTY), or [online](#). For further information regarding SDR, please visit the Office of Student Disability Resources website. In order to receive consideration for course accommodations, you must contact SDR and provide documentation (see the [documentation guidelines](#)). If the documentation supports the need for academic adjustments, SDR will provide a letter identifying appropriate academic adjustments. Please share this letter and discuss the adjustments with your instructor as early in the course as possible. You must contact SDR and request academic adjustment letters at the beginning of each semester.

Helping you succeed: The Eberly College of Science is committed to the academic success of students enrolled in the College's degree programs. Find links to help resources at <http://science.psu.edu/current-students/support-network/advising> and <http://science.psu.edu/current-students/support-network/learning-support>

Mutual Respect: ["The Eberly College of Science Code of Mutual Respect and Cooperation"](#) embodies the values that we hope our faculty, staff, and students possess and will endorse to make. The Eberly College of Science is a place where every individual feels respected and valued, as well as challenged and rewarded.